110 mm

10 mm 10 mm **Fronte**

10 mm 10 mm 34 mm 12 mm 10 mm 34 mm 12 mm xx mm

Update the duration of treatment for Ravida combination therapy in patients without cirrhosis from 12 weeks to 8 weeks.

pharmaniaga®

RAVIDA®

Tablet

200mg

COMPOSITION
Each tablet contains 219 mg ravidasvir hydrochloride equivalent to 200 mg ravidasvir.

List of excipients
Microcrystalline cellulose
Croscarmellose sodium
Colloidal silicon dioxide
Magnesium stearate
Isomalt

Excipient(s) with known effect
Each tablet contains 152.8 mg isomalt. Patients with rare hereditary problems of fructose intolerance should not take this medicine.

DESCRIPTION
White to off-white oval biconvex tablet.

INDICATIONS
Ravida is indicated in combination with other medicinal products for the treatment of chronic hepatitis C virus (HCV) infection in adults (see sections Recommended Dosage, Warnings and Precautions and Pharmacodynamics).

For HCV genotype-specific activity, see sections Warnings and Precautions and Pharmacodynamics.

RECOMMENDED DOSAGE
Treatment with Ravida should be initiated and monitored by a physician experienced in the management of patients with chronic hepatitis C.

Posology
Adults
The recommended dose of Ravida is 200 mg once daily, to be taken orally with or without food (see section Pharmacokinetics).

Ravida must be used in combination with other medicinal products indicated for the treatment of chronic hepatitis C. The package insert for the co-administered medicinal products should be referred to before initiation of therapy with Ravida.

Table 1: Recommended treatment for Ravida combination therapy in patients with compensated cirrhosis and without cirrhosis

Patient Population	Regimen and duration
Patients without cirrhosis	Ravida + sofosbuvir for 8 weeks
Patients with compensated cirrhosis	Ravida + sofosbuvir for 24 weeks

For HCV genotype information, see section Warnings and Precautions.

Dose interruption and discontinuation
If treatment interruption or discontinuation of the other medicinal products in the regimen is necessary, Ravida must not be given as monotherapy.

Vomiting and missed doses
Patients should be instructed that if vomiting occurs within 2 hours of dosing, an additional tablet should be taken. If vomiting occurs more than 2 hours after dosing, no additional dose for the day is needed. These recommendations are based on the absorption kinetics of ravidasvir, which are suggestive that the majority of the dose is absorbed within 2 hours after dosing.

Patients should be instructed that, if they miss a dose of Ravida, the dose should be taken as soon as possible if remembered within 18 hours of the scheduled dose time. However, if the missed dose is remembered more than 18 hours after the scheduled dose, the dose should be skipped and the next dose taken at the appropriate time.

Special populations
Elderly
There is limited clinical experience with Ravida in patients aged ≥ 65 years. Treatment with Ravida in accordance with the recommended posology should be guided by an assessment of the potential benefits and risks for the individual patient.

Renal impairment
No studies have been conducted with Ravida in patients with serum creatinine > 1.5 × ULN or end stage renal disease. Treatment with Ravida in accordance with the recommended posology should be guided by an assessment of the potential benefits and risks for the individual patient.

Hepatic impairment
No dose adjustment of Ravida is required for patients with mild hepatic impairment. The safety and efficacy of Ravida have been assessed in patients with compensated cirrhosis but not in patients with decompensated cirrhosis. The safety and efficacy of Ravida administration in patients with severe hepatic impairment have not been assessed (see sections Warnings and Precautions, Adverse Effects and Pharmacokinetics).

Liver transplant patients
The safety and efficacy of Ravida in the treatment of HCV infection in patients who are post-liver transplant have not been assessed. Treatment with Ravida in accordance with the recommended posology should be guided by an assessment of the potential benefits and risks for the individual patient.

Patients co-infected with human immunodeficiency virus (HIV)
The safety and efficacy of Ravida in this population have been assessed and no significant differences were observed when compared to patients not co-infected with HIV. For dosing recommendations for HIV antiviral agents co-administered with Ravida, see section Interactions with Other Medicaments.

Paediatric population
The safety and efficacy of Ravida in children and adolescents aged below 18 years have not yet been assessed.

Method of administration
Ravida is to be taken orally with or without food. Patients should be instructed to swallow the tablets whole. The tablets should not be chewed, crushed or split.

ROUTE OF ADMINISTRATION
Oral.

CONTRAINDICATIONS
Hypersensitivity to the active substance or to any of the excipients listed in section Composition.

Co-administration with potent inducers of P-glycoprotein transporter (P-gp) (e.g. carbamazepine, phenobarbital, phenytoin, rifampicin, rifabutin and St. John's wort) will decrease ravidasvir plasma concentrations and could result in loss of efficacy of Ravida (see section Interactions with Other Medicaments).

WARNINGS AND PRECAUTIONS
Ravida must not be administered as monotherapy. Ravida must be administered in combination with other medicinal products for the treatment of chronic HCV infection (see sections Indications and Recommended Dosage).

If the other associated medicinal products used in combination with Ravida are permanently discontinued, Ravida must not be given as monotherapy (see section Recommended Dosage).

Consult the package insert for the other co-administered medicinal products before starting therapy with Ravida.

Genotype-specific activity
Concerning genotype-specific virological activity and clinical outcomes, see section Pharmacodynamics. In the pivotal clinical study (STORM-C-1), the majority of patients were infected with HCV genotypes 1 or 3. There are limited data from this study in patients with HCV genotypes 2 and 6 and no data on patients with genotype 4 or 5 (see section Pharmacodynamics).

HCV/HBV (hepatitis B virus) co-infection
Cases of hepatitis B virus (HBV) reactivation, some of them fatal, have been reported during or after treatment with direct-acting antiviral agents (DAAs). HBV screening should be performed in all patients before initiation of treatment. HBV/HCV co-infected patients are at risk of HBV reactivation, and should therefore be monitored and managed according to current clinical guidelines.

Patients who have previously failed therapy with an NS5A-containing regimen
The safety and efficacy of Ravida in patients who have previously failed treatment with an NS5A-containing regimen have not been assessed.

Pregnancy and contraception requirements
Ravida is not recommended during pregnancy based on teratogenic and/or embryotoxic effects in rabbits exposed to ravidasvir. There are very limited data on exposure during pregnancy in humans.

Women of childbearing potential should use effective contraception methods before starting treatment with Ravida and continue until 1 month after completion of Ravida treatment (see section Pregnancy and Lactation).

Breast-feeding
Ravida is not recommended during breast-feeding based on the evidence that ravidasvir is able to pass through the blood-milk barrier in rats and the absence of data in human.

Interactions with medicinal products
Co-administration of Ravida can alter the concentration of other medicinal products and other medicinal products may alter the concentration of ravidasvir. Refer to section Contraindications for a listing of medicinal products that are contraindicated for use with Ravida due to potential loss of therapeutic effect. Co-administration with moderate P-gp inducers (e.g. oxcarbazepine or rifapentine) is not recommended since it may decrease ravidasvir plasma concentration leading to reduced therapeutic effect of Ravida (see section Interactions with Other Medicaments).

Use in diabetic patients
Diabetics may experience improved glucose control, potentially resulting in symptomatic hypoglycaemia, after initiating HCV DAA treatment. The physician in charge of the diabetic care of the patient should be informed when treatment with Ravida is initiated.

Paediatric population
Ravida is not recommended for use in children and adolescents aged below 18 years as the safety and efficacy have not been assessed in this population (see sections Adverse Effects and Pharmacodynamics).

Decompensated cirrhosis
The safety and efficacy of Ravida have not been assessed in patients with decompensated cirrhosis.

Hepatic impairment
No dose adjustment of Ravida is required for patients with mild hepatic impairment. The safety and efficacy of Ravida have been assessed in patients with compensated cirrhosis but not in patients with decompensated cirrhosis. The safety and efficacy of Ravida administration in patients with severe hepatic impairment has not been assessed (see sections Adverse Effects and Pharmacokinetics).

Liver transplant patients
The safety and efficacy of Ravida in the treatment of HCV infection in patients who are post-liver transplant have not been assessed.

Important information about some of the ingredients in Ravida
This medicinal product contains isomalt. Due to the presence of isomalt in the formulation, patients with rare hereditary problems of fructose intolerance should not take Ravida.

Low sodium diet information
This medicinal product contains less than 1 mmol sodium (23 mg) per dose and is considered as essentially 'sodium-free'.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES
Patients should be informed that dizziness, lethargy, somnolence and blurred vision have been reported during treatment with Ravida. No studies on the effects of Ravida on the ability to drive and use machines have been performed. Patients treated with Ravida should be warned not to drive or use machines if they feel tired or dizzy.

INTERACTIONS WITH OTHER MEDICAMENTS
Potential for ravidasvir to affect other medicinal products
Ravidasvir is an inhibitor of drug transporter breast cancer resistance protein (BCRP). Co-administration of ravidasvir with medicinal products that are BCRP substrates may increase the exposure of such medicinal products. See Table 2 for examples of interactions with sensitive substrates of BCRP (e.g. rosuvastatin).

Ravidasvir is a weak inhibitor of organic anion transporting polypeptide (OATP) 1B1 and 1B3 and of CYP3A, CYP2C19, and UGT1A1. Clinically meaningful interactions are not expected between Ravida and substrates of BCRP, OATP, CYP enzymes or UGT1A1.

Potential for other medicinal products to affect ravidasvir
Ravidasvir is a substrate of the drug transporter P-glycoprotein (P-gp). Medicinal products that are strong inducers of P-gp (e.g. carbamazepine, phenobarbital, phenytoin, rifampicin, rifabutin and St. John's wort) may decrease plasma concentrations of ravidasvir leading to reduced therapeutic effect of Ravida. The use of such medicinal products with Ravida is contraindicated (see section Contraindications).

Medicinal products that are moderate P-gp inducers (e.g. oxcarbazepine or rifapentine) may decrease ravidasvir plasma concentration leading to reduced therapeutic effect of Ravida. Co-administration of Ravida with moderate P-gp inducers is not recommended (see section Warnings and Precautions).

Co-administration with medicinal products that inhibit P-gp may increase ravidasvir plasma concentrations. However, clinically meaningful interactions with P-gp inhibitors are not expected.

Patients treated with vitamin K antagonists
As liver function may change during treatment with ravidasvir, close monitoring of International Normalised Ratio (INR) values is recommended.

Interactions between Ravida and other medicinal products
Table 2 provides a listing of established or potentially clinically relevant medicinal product interactions. The interactions described are based on studies conducted with ravidasvir or are predicted medicinal product interactions that may occur with ravidasvir. The direction of the arrow indicates the direction of change of the exposures. Observed or predicted changes are shown as ↑ = increase, ↓ = decrease, ↔ = no change. The table is not all-inclusive.

Table 2: Interaction between Ravida and other medicinal products

Medicinal products by therapeutic areas (Possible Mechanism of Interaction)	Interaction	Recommendation concerning co-administration with ravidasvir
ANGIOTENSIN II RECEPTOR BLOCKERS		
Valsartan	Interaction not studied. Expected: ↔ Ravidasvir ↔ Valsartan	No dose adjustment of Ravida or valsartan is required.
Losartan	Interaction not studied. Expected: ↔ Ravidasvir ↔ Losartan	No dose adjustment of Ravida or losartan is required.
ACID REDUCING AGENTS		
		Solubility of ravidasvir decreases as pH increases. Medicinal products that increase gastric pH are expected to decrease the concentration of ravidasvir.
Antacids		
e.g. aluminium or magnesium hydroxide; calcium carbonate (Increase in gastric pH)	Interaction not studied. Expected: ↓ Ravidasvir ↔ Antacid	It is recommended to separate antacid use and Ravida administration by 4 hours.
H2-receptor antagonists		
Famotidine (Increase in gastric pH)	Interaction not studied. Expected: ↓ Ravidasvir ↔ Famotidine	It is recommended that H2-receptor antagonists should be administered 4 hours after Ravida.
Cimetidine Nizatidine Ranitidine (Increase in gastric pH)	Interaction not studied. Expected: ↓ Ravidasvir ↔ H2-receptor antagonist	Same class as famotidine and similar interactions predicted.
Proton pump inhibitors		
Omeprazole (Increase in gastric pH)	Interaction studied. Observed: ↔ Ravidasvir ↔ Omeprazole	No dose adjustment of Ravida or omeprazole is required.
Esomeprasole Lansoprazole Pantoprazole Rabeprazole (Increase in gastric pH)	Interaction not studied. Expected: ↔ Ravidasvir ↔ Proton pump inhibitor	Same class as omeprazole and similar interactions predicted.
ANTIARRHYTHMICS		
Digoxin	Interaction not studied. Expected: ↔ Ravidasvir ↔ Digoxin	No dose adjustment of Ravida or digoxin is required. Caution is warranted because of the narrow therapeutic window of digoxin.
ANTICOAGULANTS		
Dabigatran etexilate	Interaction not studied. Expected: ↔ Ravidasvir ↔ Dabigatran etexilate	No dose adjustment of Ravida or dabigatran etexilate is required. Caution is warranted because of the narrow therapeutic window of dabigatran etexilate.
Vitamin K antagonists	Interaction not studied.	Close monitoring of INR is recommended with all vitamin K antagonists. This is due to liver function changes during treatment with Ravida.
ANTICONVULSANTS		
Carbamazepine Phenytoin Phenobarbital (Induction of P-gp)	Interaction not studied. Expected: ↓ Ravidasvir	Ravida is contraindicated with carbamazepine, phenobarbital and phenytoin, potent P-gp inducers (see section Contraindications).
Oxcarbazepine (Induction of P-gp)	Interaction not studied. Expected: ↓ Ravidasvir	Co-administration of Ravida with oxcarbazepine is not recommended.
ANTIDIABETICS		
Metformin (Inhibition of P-gp)	Interaction not studied. Expected: ↑ Ravidasvir ↔ Metformin	No dose adjustment of Ravida or metformin is required.
ANTIFUNGALS		
Ketoconazole Itraconazole Posaconazole Isavuconazole (Inhibition of P-gp)	Interaction not studied. Expected: ↑ Ravidasvir ↔ Ketoconazole ↔ Itraconazole ↔ Posaconazole ↔ Isavuconazole	No dose adjustment of Ravida, ketoconazole, itraconazole, posaconazole or isavuconazole is required.
Voriconazole Fluconazole	Interaction not studied. Expected: ↔ Ravidasvir ↔ Voriconazole ↔ Fluconazole	No dose adjustment of Ravida, voriconazole or fluconazole is required.
ANTIMYCOBACTERIALS		
Rifampicin (Induction of P-gp)	Interaction not studied. Expected: ↓ Ravidasvir ↔ Rifampicin	Ravida is contraindicated with rifampicin, a potent P-gp inducer (see section Contraindications).
Rifabutin (Induction of P-gp)	Interaction not studied. Expected: ↓ Ravidasvir ↔ Rifabutin	Ravida is contraindicated with rifabutin, a potent P-gp inducer (see section Contraindications).
Rifapentine (Induction of P-gp)	Interaction not studied. Expected: ↓ Ravidasvir ↔ Rifapentine	Co-administration of ravidasvir with rifapentin is not recommended.
BENZODIAZEPINES		
Midazolam	Interaction studied Observed: ↔ Midazolam Expected: ↔ Ravidasvir	No dose adjustment of Ravida or midazolam is required.
HIV ANTIVIRAL AGENTS: REVERSE TRANSCRIPTASE INHIBITORS		
Tenofovir disoproxil fumarate	Interaction studied. Observed: ↔ Tenofovir Expected: ↔ Ravidasvir	No dose adjustment of Ravida or tenofovir disoproxil fumarate is required.
Lamivudine	Interaction not studied. Expected: ↔ Ravidasvir ↔ Lamivudine	No dose adjustment of Ravida or lamivudine is required.
Zidovudine	Interaction not studied. Expected: ↔ Ravidasvir ↔ Zidovudine	No dose adjustment of Ravida or zidovudine is required.
Emtricitabine	Interaction studied: Observed: ↔ Emtricitabine Expected: ↔ Ravidasvir	No dose adjustment of Ravida or emtricitabine is required.
Abacavir	Interaction not studied. Expected: ↔ Ravidasvir ↔ Abacavir	No dose adjustment of Ravida or abacavir is required.
Stavudine	Interaction not studied. Expected: ↔ Ravidasvir ↔ Stavudine	No dose adjustment of Ravida or stavudine is required.
Efavirenz	Interaction studied. Observed: ↔ Efavirenz Expected: ↔ Ravidasvir	No dose adjustment of Ravida or efavirenz is required.
Nevirapine	Interaction studied. Observed: ↔ Nevirapine Expected: ↔ Ravidasvir	No dose adjustment of Ravida or nevirapine is required.
HIV ANTIVIRAL AGENTS: PHARMACOKINETIC (PK) BOOSTERS		
Cobicistat (P-gp inhibitor)	Interaction not studied. Expected: ↑ Ravidasvir ↔ Cobicistat	No dose adjustment of Ravida or cobicistat is required.
Ritonavir	Interaction not studied. Expected: ↔ Ravidasvir ↔ Ritonavir	No dose adjustment of Ravida or ritonavir is required.
HIV ANTIVIRAL AGENTS: HIV PROTEASE INHIBITORS		
Atazanavir (P-gp inhibitor)	Interaction not studied. Expected: ↑ Ravidasvir ↔ Atazanavir	No dose adjustment of Ravida or atazanavir is required.
Lopinavir (P-gp inhibitor)	Interaction not studied. Expected: ↑ Ravidasvir ↔ Lopinavir	No dose adjustment of Ravida or lopinavir is required.
Darunavir (P-gp inhibitor)	Interaction not studied. Expected: ↑ Ravidasvir ↔ Darunavir	No dose adjustment of Ravida or darunavir is required.
HIV ANTIVIRAL AGENTS: INTEGRASE INHIBITORS		
Raltegravir	Interaction not studied. Expected: ↔ Ravidasvir ↔ Raltegravir	No dose adjustment of Ravida or raltegravir is required.
Elvitegravir (P-gp inhibitor)	Interaction not studied. Expected: ↑ Ravidasvir ↔ Elvitegravir	No dose adjustment of Ravida or elvitegravir is required.
Dolutegravir	Interaction not studied. Expected: ↔ Ravidasvir ↔ Dolutegravir	No dose adjustment of Ravida or dolutegravir is required.
HERBAL SUPPLEMENTS		
St. John's wort (Induction of P-gp)	Interaction not studied. Expected: ↓ Ravidasvir	Ravida is contraindicated with St John's wort, a potent P-gp inducer (see section Contraindications).
HMG-CoA REDUCTASE INHIBITORS		
Pravastatin	Interaction not studied. Expected: ↔ Ravidasvir ↔ Pravastatin	No dose adjustment of Ravida or pravastatin is required. Caution is recommended during the combination treatment.
Pitavastatin	Interaction not studied. Expected: ↔ Ravidasvir ↔ Pitavastatin	No dose adjustment of Ravida or pitavastatin is required. Caution is recommended during the combination treatment.
Rosuvastatin, Simvastatin, Atorvastatin, Fluvastatin (inhibition of BCRP)	Interaction not studied. Expected: ↔ Ravidasvir ↑ Rosuvastatin ↑ Simvastatin ↑ Atorvastatin ↑ Fluvastatin	Co-administration of Ravida with rosuvastatin, simvastatin, atorvastatin or fluvastatin could increase the concentration of these drugs, which is associated with increased risk of myopathy, including rhabdomyolysis. Monitoring for possible adverse effects is recommended. A low dose of the statin may be considered at the initiation of the DAA treatment.
NARCOTIC ANALGESICS		
Methadone	Interaction not studied. Expected: ↔ Ravidasvir ↔ Methadone	No dose adjustment of Ravida or methadone is required.
Buprenorphine	Interaction not studied. Expected: ↔ Ravidasvir ↔ Buprenorphine	No dose adjustment of Ravida or buprenorphine is required.
IMMUNOSUPPRESSANTS		
Ciclosporin (P-gp inhibitor)	Interaction not studied. Expected: ↑ Ravidasvir ↔ Ciclosporin	No dose adjustment of Ravida or ciclosporin is required.
Tacrolimus	Interaction not studied. Expected: ↔ Ravidasvir ↔ Tacrolimus	No dose adjustment of Ravida or tacrolimus is required.
ORAL CONTRACEPTIVES		
Norgestimate/ ethinyl estradiol	Interaction not studied. Expected: ↔ Norelgestromin ↔ Norgestrel ↔ Ethinyl estradiol	No dose adjustment of Ravida or norgestimate/ethinyl estradiol is required.

PREGNANCY AND LACTATION
Fertility
No human data on the effect of ravidasvir on fertility are available. No effect on fertility was seen in rats (see section Preclinical Safety Data).

Pregnancy
Teratogenic and/or embryotoxic effects have been demonstrated in rabbits exposed to ravidasvir. There are very limited data on exposure to Ravida during pregnancy in humans. Use of Ravida is not recommended during pregnancy.

Women of childbearing potential should use effective contraception methods before starting treatment with Ravida and continue until 1 month after completion of treatment (see section Preclinical Safety Data).

Breast-feeding
Ravidasvir has been shown to pass through the blood-milk barrier in rats (see section Preclinical Safety Data). There are no data in humans. Use of Ravida is not recommended during breast-feeding.

ADVERSE EFFECTS
Summary of the safety profile
The summary of the safety profile of Ravida is based on data from 301 adult subjects with chronic HCV infection (with no or compensated cirrhosis), who received Ravida in combination with sofosbuvir once daily in the STORM-C-1 study. Among them, 90 were HIV co-infected.

Adverse reactions described below relate to treatment with the combination of Ravida plus sofosbuvir. No adverse reactions specific to Ravida alone have been identified.

Ravida in combination with sofosbuvir
The majority of adverse reactions identified with Ravida co-administered with sofosbuvir were of mild or moderate intensity, regardless of the presence or absence of cirrhosis, HIV co-infection, prior HCV treatment or HCV genotype. There were few severe adverse reactions and none that were life-threatening. Two subjects discontinued the Ravida plus sofosbuvir regimen due to adverse events assessed as being related to study treatment.

The most frequent adverse reactions identified in patients receiving Ravida and sofosbuvir were pyrexia (11.6%), headache (6.6%), dizziness (5.6%), lethargy (5.3%), dyspepsia (5.0%) and nausea (5.0%).

Tabulated list of adverse reactions
Based on STORM-C-1 data in 301 subjects, adverse reactions identified with Ravida administered in combination with sofosbuvir are listed in Table 3 by MedDRA System Organ Class, with Preferred Term (PT).

Frequencies are defined as follows: very common (≥1/10), common (≥1/100 to <1/10), uncommon (≥1/1,000 to <1/100), rare (≥1/10,000 to <1/1,000) and very rare (<1/10,000).

Table 3: Adverse reactions reported in ≥ 2 patients by decreasing frequency

System Organ Class	Adverse Reactions Preferred Term *Ravida + sofosbuvir* N = 301		
	Very common	Common	Uncommon
Metabolism and nutrition disorders			decreased appetite
Psychiatric disorders		insomnia	irritability
Nervous system disorders		headache; dizziness; lethargy; somnolence	
Cardiac disorders			palpitations
Vascular disorders			hot flush
Gastrointestinal disorders		dyspepsia; nausea; diarrhoea; abdominal pain; vomiting; constipation	
Skin and subcutaneous tissue disorders		pruritus, rash; dermatitis	
General disorders and administration site conditions	pyrexia	fatigue	
Investigations		aspartate aminotransferase increased	International Normalised Ratio increased

Description of selected adverse reactions
HCV/HIV co-infection
The overall safety profile of Ravida administered in combination with sofosbuvir in HCV/HIV co-infected subjects was comparable to that observed in HCV mono-infected subjects.

General recommendation on combination therapy:
Refer to the package insert of other medicinal products with which Ravida will be combined for descriptions of selected adverse reactions.

Reporting of suspected adverse reactions
Reporting suspected adverse reactions after registration of the medicinal product is important. It allows continued monitoring of the safety profile and benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the national reporting system.

OVERDOSE AND TREATMENT
The highest documented dose of Ravida was 320 mg once daily for 5 days in healthy volunteers. There were no untoward effects observed at these dose levels and adverse events were similar in frequency and severity to those reported in the placebo groups. The effects of higher doses/exposures are not known.

No specific antidote is available for overdose with ravidasvir. If overdose occurs, the patient must be monitored for evidence of toxicity. Treatment of overdose with Ravida consists of general supportive measures including monitoring of vital signs, as well as observation of the clinical status of the patient.

PHARMACODYNAMICS
Pharmacotherapeutic group: Antivirals for systemic use, direct-acting antivirals (DAAs).

Mechanism of action
Ravidasvir is an inhibitor of non-structural protein 5A (NS5A), a multifunctional protein that is an essential component of the HCV replication complex. Ravidasvir inhibits replication of variant HCV replicons encoding resistance mutations for the other major classes of HCV DAAs. HCV variants with reduced susceptibility to ravidasvir remain fully susceptible to other classes of HCV inhibitors.

Antiviral activity
The 50% effective concentration (EC_{50}) values and 90% effective concentration (EC_{90}) values of ravidasvir for various HCV genotypes 1a and 1b and in non-genotype 1 replicon cells are presented in Table 4.

Table 4: Ravidasvir Susceptibility in HCV GT1a and GT1b and in non-GT1 Replicon Cells

HCV Genotype	EC_{50} ± SD		EC_{90} ± SD	
	nM	ng/mL	nM	ng/mL
1a	0.12 ± 0.01	0.09	0.32 ± 0.02	0.24
1b	0.02 ± 0.002	0.015	0.03 ± 0.004	0.02
2a NS5A	0.14 ± 0.03	0.12 ± 0.03	0.45 ± 0.1	0.38 ± 0.08
3a NS5A	1.14 ± 0.3	0.95 ± 0.25	3.35 ± 1.3	2.80 ± 1.09
4a NS5A	0.05 ± 0.01	0.04 ± 0.01	0.13 ± 0.015	0.11 ± 0.01
5a NS5A	0.04 ± 0.01	0.03 ± 0.01	0.12 ± 0.02	0.10 ± 0.02
6a NS5A	0.30 ± 0.1	0.25 ± 0.08	1.1 ± 0.3	0.920 ± 0.25
7a NS5A	0.07 ± 0.02	0.06 ± 0.02	0.2 ± 0.05	0.17 ± 0.04

Abbreviations: EC_{50} = effective concentration reducing viral replication by 50%; EC_{90} = effective concentration reducing viral replication by 90%; HCV = hepatitis C virus; SD = standard deviation.

Increasing the concentration of human serum from 1% to 40% led to a 9- to 11-fold shift in EC_{50} and EC_{90} values for genotypes 1a and 1b, consistent with the high protein binding observed in equilibrium dialysis studies.

10 mm

Combination index values calculated for paired inhibitors indicate additive effects when ravidasvir is combined with NS3 protease inhibitor (ITMN-191), interferon-α and non-nucleoside polymerase inhibitor (HCV-796) and additive to synergistic effects when combined with a NS5B nucleoside polymerase inhibitor (PSI-6130). No antagonism was observed.

No significant inhibitory activity was observed against a panel of RNA and DNA viruses such as bovine viral diarrheal virus (BVDV), human rhino virus (HRV)-16 and influenza-B (Flu-B), confirming that ravidasvir is a specific inhibitor of HCV replication.

Resistance

In cell cultures

Replicons of HCV genotypes 1a and 1b and a panel of genotype 1b chimera replicons containing critical segments of the NS5A gene of HCV genotypes 2a, 3a, 4a, 5a and 6a have been evaluated *in vitro* with the highest concentration capped at 1,000 nM ravidasvir. Resistant colonies emerged on most plates, with the number of resistant colonies being lower at the higher concentrations of ravidasvir. At a ravidasvir concentration of 1,000 nM only a small number of colonies were observed with HCV genotype 2a and 4a NS5A replicon cells. Genotyping of replicons from recovered colonies confirmed the presence of resistance substitutions.

Consistent with patterns of resistance-associated substitutions for other NS5A inhibitors, one or more changes at amino acid positions 28, 30, 31 and 93 within domain 1 of the NS5A protein were observed. Additional substitutions, such as P58L (4a NS5A) and T58A (6a NS5A), emerged at lower concentrations of ravidasvir, but these were not present at high ravidasvir concentrations (1,000 nM). Colony formation assays using HCV genotype 1a replicon cells yielded the most diverse set of substitutions, suggesting that multiple single and linked substitutions confer resistance in this genotype. This is in contrast with the results obtained using the HCV genotype 2a, 3a, and 6a NS5A replicon cell lines, which consistently yielded single preferred substitutions F28S (genotypes 2a and 6a) and Y93H (genotype 3a) that conferred high-level resistance. Similar to the linked substitutions found in genotype 1a replicon cells, combination substitutions L31F/V+Y93H (genotype 1b), L30H+Y93H (genotype 4a) and Q30H+L31F (genotype 5a) are required to achieve high level resistance.

In clinical studies

Ravida monotherapy

Three days of Ravida monotherapy at doses ranging from 40 mg to 240 mg once daily led to a rapid reduction of HCV RNA level in 35/40 subjects. Seven of the 35 had NS5A resistance associated variants (RAVs) at baseline. The 5 non-responders, who had <1 $\log_{10}$ reduction in HCV RNA on day 4, included one genotype 1b subject with genetically linked R30Q+L31I+Y93H RAVs, one genotype 2a with F28L+L31M, two genotype 3A with Y93H and one genotype 3a with A30K.

As wild-type virus was eliminated, treatment-emergent RAVs were detected in 31/32 genotype 1 subjects 8-48 hours after first dose, however these were not further enriched with continued treatment, indicating that Ravida can inhibit replication of these resistant variants. The percentage of linked substitutions was low overall in all genotype 1 subjects, consistent with enrichment of pre-existing variants rather than emergence of new strains.

The most common emergent RAVs in genotype 1a subjects were M28T, Q30R, Q30H, L31M, Y93H and Y93C, with Q30R the most frequent. In genotype 1b subjects, R30Q, L31M and Y93H were the most common emergent RAVs, with Y93H the most frequent. Y93H was the most frequent emergent RAV in the genotype 3a subjects.

Ravida in combination with NS5B inhibitor

Viral resistance has been assessed in HCV chronically infected adults with no or compensated cirrhosis treated with Ravida plus sofosbuvir in the STORM-C-1 study. The study population included 301 subjects infected with virus genotypes 1 (42%), 2 (1%), 3 (52%) and 6 (6%). Among the 81 cirrhotic subjects, 65% had genotype 3 infection.

HCV RNA sequences were obtained for 292/301 subjects (all except 4 non-virological failures and 5 for whom sequencing failed). NS5A RAVs were detected in 91/292 (31.2%) at baseline. The distribution of the RAVs was heterogeneous across genotypes, with 28V and 93H predominantly found in the genotype 1a subjects, 93H in genotype 1b, 93H and 62L in genotype 3a; 30K and 31M in genotype 3b, and 28V and 93S in geno type 6. The impact of baseline RAVs on SVR12 is shown in Table 5.

Table 5: Impact of baseline RAVs on SVR12 in STORM-C-1

Number of baseline NS5A RAVs	SVR12 n/N (%)
Any	86/91 (94.5%)
1	53/55 (96.4%)
2	31/34 (91.2%)
3	2/2 (100%)
None	201/201 (100%)

n = number of subjects meeting specified criteria; N = number of subjects; NS5A = non-structural protein 5A; RAV = viral resistance associated variants; SVR12 = sustained virologic response 12 weeks after end of treatment.

The presence of single or multiple baseline NS5A RAVs did not impact treatment outcomes for genotypes 1a, 1b, 2 or 3b. SVR12 in subjects with one or more RAVs was 93.5% in genotype 3a and 78.6% in genotype 6. Among the 23 genotype 3a subjects with the NS5A 93H RAV, 21 (91.3%) achieved SVR12. 11 of the genotype 6 subjects, including all 3 virological failures, had the NS5A double RAV 28V+93S. 10 of these 11 were genotype 6n, which has both RAVs in the reference baseline sequence.

There was no emergence of new NS5A RAVs in any of the subjects who experienced virological failure. One genotype 6n subject had an emergent NS5B (sofosbuvir) RAV (282T/C) after virological failure.

Cross-resistance

A series of cross-resistance assays was performed and the data indicate that ravidasvir retained its ability to inhibit each of these resistant variants with EC_{50} values equivalent to those observed in the wild-type genotype 1b replicon cells, confirming the ability of ravidasvir to inhibit variants that emerge with other distinct classes of HCV inhibitors. Ravidasvir-resistant variants retained their susceptibility to other classes of HCV inhibitors. Ravidasvir can be used in combination with other classes of HCV inhibitors to help prevent the emergence of resistant strains.

Clinical efficacy and safety

Ravidasvir + NS5B inhibitor

The efficacy and safety of Ravida 200 mg QD (once daily) + sofosbuvir 400 mg QD was evaluated in the pivotal study STORM-C-1 conducted in Malaysia and Thailand. The primary efficacy endpoint of STORM-C-1 was sustained virologic response 12 weeks after end of treatment (SVR12) as determined by HCV RNA level <LLOQ using a real-time polymerase chain reaction (PCR) method (Roche COBAS AmpliPrep/TaqMan or Abbott m2000 system).

Of the 301 subjects enrolled in STORM-C-1, 220 non-cirrhotic subjects were allocated to Ravida + sofosbuvir for 12 weeks and 81 compensated cirrhotic subjects were allocated to Ravida + sofosbuvir for 24 weeks. A total of 298 (99%) subjects completed the study and 3 (1%) subjects (2 non-cirrhotic and 1 cirrhotic) prematurely discontinued the study due to adverse events. Demographic and baseline characteristics are summarised in Table 6.

Table 6: Demographic and Baseline Characteristics (STORM-C-1)

Parameter	Category	RDV + SOF 12 weeks (non cirrhotic) N=220	RDV + SOF 24 weeks (cirrhotic) N=81	Overall N=301
Sex	Male	170 (77%)	61 (75%)	231 (77%)
	Female	50 (23%)	20 (25%)	70 (23%)
Age (years)	Median (Q1, Q3)	44 (37, 52)	53 (46, 57)	47 (40, 55)
Ethnicity	Malay	74 (34%)	46 (57%)	120 (40%)
	Thai	79 (36%)	0	79 (26%)
	Chinese	48 (22%)	30 (37%)	78 (26%)
	Other	19 (8%)	5 (6%)	24 (8%)
Race	Asian	220 (100%)	81 (100%)	301 (100%)
FibroScan result (kPa)	Median (Q1, Q3)	6.9 (5.5, 8.9)	21.3 (16.9, 27.7)	8.6 (6.1, 14.0)
Child-Turcotte-Pugh score[1]	5	-	71 (88%)	-
	6	-	10 (12%)	-
Hepatitis C genotype	GT1	97 (44%)	28 (35%)	125 (42%)
	GT2	2 (1%)	0	2 (1%)
	GT3	105 (48%)	53 (65%)	158 (52%)
	GT6	16 (7%)	0	16 (5%)
IL28B	CC	162 (74%)	60 (74%)	222 (74%)
	CT	58 (26%)	20 (25%)	78 (26%)
	TT	0	1 (1%)	1 (<1%)
HCV RNA ($\log_{10}$ IU/mL)	Median (Q1, Q3)	6.4 (5.8, 6.8)	6.2 (5.8, 6.6)	6.3 (5.8, 6.7)
HCV RNA (IU/mL)	≥800,000	158 (72%)	57 (70%)	215 (71%)
ALT (U/L)	Median (Q1, Q3)	59 (39, 92)	100 (70, 157)	71 (43, 111)
AST (U/L)	Median (Q1, Q3)	47 (33, 66)	86 (59, 113)	55 (37, 81)
HIV serology	Positive	79 (36%)	11 (14%)	90 (30%)
HIV-1 RNA viral load (copies/mL)	<20	74 (94%)	7 (64%)	81 (90%)
	≥20 - <1000	5 (6%)	4 (36%)	9 (10%)

Safety analysis set, n=301; ALT = alanine aminotransferase; AST = aspartate aminotransferase; HCV = hepatitis C virus; HIV = human immunodeficiency virus; IL28 = interleukin 28; Max = maximum; Min = minimum; N = number of subjects; RNA = ribonucleic acid; SD = standard deviation; SOF = sofosbuvir; Q1 = quartile 1; Q3 = quartile 3;

[1] Only cirrhotic subjects

[2] Only subjects co-infected with HIV

High SVR12 results above 95% were observed in the key subpopulations including cirrhotic (96.3%) and non-cirrhotic subjects (97.3%), HIV co-infected (96.7%) or not (97.1%), previous IFN experienced (96.0%) or not (97.5%), and across genotype 1a (99.0%), genotype 1b (100.0%), and genotype 3a (96.4%). Genotype 3 cirrhotic subjects achieved 96.8% SVR12. Subjects with genotype 6 achieved 81.3% SVR12 (13 of 16 subjects). Both genotype 2 subjects achieved SVR12.

No significant association was observed between SVR12 and sex, age, ethnicity (Malay, Thai, other), site, baseline BMI, cirrhotic status, baseline HCV RNA, baseline ALT, baseline AST, IL28B, prior interferon-based HCV treatment status, or HIV co-infection status (p>0.05).

Treatment with Ravida + sofosbuvir led to a rapid decrease in viral load. The mean HCV ribonucleic acid (RNA) count was 6.15 $\log_{10}$ IU/mL at baseline and decreased to 2.03 $\log_{10}$ IU/mL at week 1, 1.18 $\log_{10}$ IU/mL at week 4, 1.14 $\log_{10}$ IU/mL at week 8, and 1.12 $\log_{10}$ IU/mL at week 12.

Table 7 shows summary of the main efficacy results.

Overall, 9 subjects (6 [3%] non-cirrhotic subjects and 3 [4%] cirrhotic subjects) did not achieve SVR12; 1 subject due to virological breakthrough, 4 subjects due to virological relapse, and 4 subjects due to non-virologic failure (Table 8). Genotype 3 status, HIV status, and prior interferon-based HCV treatment status did not appear to affect SVR12 rate. Three virological failures were observed in the genotype 6 group. No other risk factors were identified among the subjects who failed treatment.

Table 7: Summary of Virologic Outcomes in STORM-C-1

Study and Group	Dosing regimen	SVR4	SVR12	SVR24
Overall (n=300)	Both regimens as described below	97.7% (95.3-99.1%)	97.0% (94.4-98.6%)	96.3% (93.5-98.2%)
Non-cirrhotic subjects (n=219)	Ravida + sofosbuvir 12 weeks	98.2% (95.4-99.5%)	97.3% (94.1- 99.0%)	96.3% (92.9-98.4%)
Cirrhotic subjects (n=81)	Ravida + sofosbuvir 24 weeks	96.3% (89.6-99.2%)	96.3% (89.6-99.2%)	96.3% (89.6-99.2%)

Full analysis set, n=300

Table 8: Virologic Outcomes at Follow up Week 12 Visit by Key Subgroup (STORM-C-1)

Parameter	No cirrhosis N=219	Cirrhosis N=81	Genotype 3 N=158	Non 3 genotype N=142	HIV co-infection N=90	No HIV co infection N=210	Prior interferon - based HCV treatment N=99	No prior interferon-based HCV treatment N=201	Overall N=300
Subjects achieving SVR12	213 (97%)	78 (96%)	153 (97%)	138 (97%)	87 (97%)	204 (97%)	95 (96%)	196 (98%)	291 (97.0%)
Subjects not achieving SVR12	6 (3%)	3 (4%)	5 (3%)	4 (3%)	3 (3%)	6 (3%)	4 (4%)	5 (2%)	9 (3.0%)
Virologic failure	4 (2%)	1 (1%)	2 (1%)	3 (2%)	1 (1%)	4 (2%)	2 (2%)	3 (1%)	5 (1.7%)
On treatment virologic failure	0	1 (1%)	1 (1%)	0	0	1 (<1%)	1 (1%)	0	1 (0.3%)
Virologic breakthrough	0	1 (1%)	1 (1%)	0	0	1 (<1%)	1 (1%)	0	1 (0.3%)
Virologic relapse	4 (2%)	0	1 (1%)	3 (2%)	1 (1%)	3 (1%)	1 (1%)	3 (1%)	4 (1.3%)
Non virologic failure	2 (1%)	2 (2%)	3 (2%)	1 (1%)	2 (2%)	2 (1%)	2 (2%)	2 (1%)	4 (1.3%)

Full analysis set, n=300; N = number of subjects

Ravidasvir + other DAAs

In published clinical studies, ravidasvir 200 mg once daily demonstrated ≥ 95% SVR12 across all tested genotypes when combined with other appropriate DAAs (Table 9).

Table 9: Published SVR12 rates observed with ravidasvir used in combination with other DAAs

Study	N	Population	Treatment	SVR12
PYRAMID (Esmat et al, J Hepatol. 2017 Sep 19. pii: S0168-8278 (17)32286-9)	300	Chronic HCV-GT4 infection, with or without compensated CTP-A cirrhosis, with serum HCV RNA levels ≥10,000 IU/ml, treatment-naïve or treatment-experienced	Group 1a, 1b, and 2 (12 weeks): • Ravidasvir 200 mg once daily + sofosbuvir 400 mg once daily, OR • Ravidasvir 200 mg once daily + sofosbuvir 400 mg once daily + ribavirin 1200 mg (≥75 kg), and 1000 mg daily (<75 kg), given in two divided doses. Group 3 (16 weeks): • Ravidasvir 200 mg once daily + sofosbuvir 400 mg once daily + ribavirin 1200 mg daily (≥75 kg), and 1000 mg daily (<75 kg), given in two divided doses.	**95.3%**
EVEREST (Kao et al, J Gastroenterol Hepatol. 2018 Aug;33(8):15 07-1510)	38	Chronic HCV GT1 infection without cirrhosis Plasma HCV RNA concentration > 10 000 IU/mL, and no prior interferon-based HCV treatment	Ravidasvir 200 mg tablets once daily Ritonavir-boosted danoprevir 100 mg/100 mg twice daily Ribavirin 500 mg (<75 kg) or 600 mg (≥75 kg) twice daily 12 weeks	**100%**
ASC-ASC16-II/III-CTP-1-01 (Xu et al, J Clin Transl Hepatol. 2019 Sep 28;7(3):213-2 20)	424	Chronic GT1 HCV infection without cirrhosis. Plasma HCV RNA ≥ 10,000 IU/mL; and no prior interferon-based HCV treatment	Ravidasvir 200 mg once daily Ritonavir-boosted danoprevir 100 mg/100 mg twice daily and oral Ribavirin at 1000 (<75 kg)/1200 (≥75 kg) mg/day for 12 weeks Placebo group: placebo for 12 weeks followed by deferred active treatment	**96% (treatment arm) 95% (deferred-treatment arm)**

An open-label, randomized, multicenter study was conducted to evaluate the efficacy and safety of the 8-week versus 12-week SOF-RDV combination treatment for non-cirrhotic chronic hepatitis C patients in the demographic, clinical, and biological characteristics of patients achieving SVR12. A total of 322 subjects were randomized, and 308 subjects completed the study. Participants are patients with all HCV genotypes, with or without HIV co-infection, regardless of the source of infection. The experimental arm received treatment for 8 weeks, while the control arm patient was 12 weeks.

The primary endpoint was the sustained virologic response at 12 weeks post-treatment (SVR12). The comparison of treatment duration was assessed using both the intent-to-treat (ITT) and per-protocol (PP) analyses. In the ITT analysis shows the SVR12 rate difference between the 8-week and 12-week groups was -0.04% (95% CI: -4.94, 4.85), as presented in Table 10. Since the lower limit of the confidence interval (-4.94) was above the non-inferiority margin (-5), the 8-week regimen was considered non-inferior to the 12-week regimen. However, in the PP analysis, the SVR12 rate difference was -0.37% (95% CI: -5.36, 4.63; p = 0.0637). The lower confidence interval limit (-5.36) fell below the non-inferiority margin (-5), indicating that non-inferiority of the 8-week treatment could not be established. Since the ITT and PP analyses produced different results, the findings were not consistent across both methods.

Table 10: Analysis of Achieved HCV-RNA Viral Load at week 12/follow-up visit – ITT Population

SVR12	Achieved	Not Achieved
8-weeks group (N = 159) n/N1 (%)	142/152 (93.42)	10/152 (6.58)
12-weeks group N = 160) n/N1 (%)	143/153 (93.46)	10/153 (6.54)
Percentage Difference (week 8 vs week 12)	-0.04	
One sided 95% CI	(-4.94, 4.85)	
p-value	0.0479	
Non-Inferiority criteria	The lower limit of one-sided 95% CI should be > NIM (-5).	
Conclusion	The lower limit of 95% CI is -4.94 which is > NIM (-5). So, 8-weeks group is non-inferior to the 12 weeks group.	

The ITT analysis showed that the mean HCV RNA levels were comparable between the two groups, with 3,030,005 IU/mL in the 8-week group and 3,328,121.5 IU/mL in the 12-week group. A higher percentage of patients in the 12-week group had a baseline viral load of <800,000 IU/mL (38.13% vs. 33.96%), while a greater percentage of patients in the 8-week group had a viral load of ≥800,000 IU/mL (66.04% vs. 61.88%). The most common genotypes were Genotype 1 (33.96% in 8-week group, 29.38% in 12-week group) and Genotype 3 (57.86% in 8-week, 61.25% in 12-week). No cases of Genotypes 4, 5, or 6 were reported.

Liver Enzyme Normalization (ALT & AST) Analysis: Liver enzyme normalization was evaluated at Week 12 post-treatment. ALT levels normalized was observed in 33.33% of patients in the 8-week group and 36.88% in the 12-week group. AST normalization occurred in 43.40% of patients in the 8-week group and 38.13% in the12-week group. Findings from safety and per-protocol (PP) analyses were consistent with ITT analysis results.

Overall, combination treatment of Sofosbuvir (SOF,Grateziano) 400 mg tablet and Ravidasvir (RDV, formerly PPI-668) 200 mg tablet for 8 weeks group is comparable with 12 weeks group in non-cirrhotic chronic hepatitis C patients. The safety profile of combination treatment of Sofosbuvir (SOF,Grateziano) 400 mg tablet and Ravidasvir (RDV, formerly PPI-668) 200 mg tablet for 8 weeks group is comparable with 12 weeks group.

PHARMACOKINETICS

The mean steady-state pharmacokinetic parameters of ravidasvir in 25 HCV mono-infected subjects receiving 200 mg ravidasvir and 400 mg sofosbuvir QD are presented in Table 10.

Table 11: Mean (CV%) steady-state ravidasvir pharmacokinetic parameters in HCV infected patients

Dose	200 mg
C_{max} (ng/mL)	2 ,540 (47.6)
T_{max}[1] (h)	2.0 (1.0-3.1)
C_{trough} (ng/mL)	190 (106.9)
AUC_{0-24} (h·ng/mL)	19,920 (64.1)
CL_{ss}/F (L/h)	13.99 (80.4)

AUC_{0-24} = area under the concentration-time curve from time 0 to 24 hours; CL_{SS}/F = steady-state apparent systemic clearance; C_{max} = maximum plasma concentration; C_{trough} = 24-hour postdose concentration; T_{max} = time to maximum concentration

[1] Median and (range) for T_{max}

Absorption

Following oral administration of ascending single doses (80-320 mg) to healthy subjects, ravidasvir was relatively rapidly absorbed with concentrations peaking at a median of 1.5 to 2 hours post-dose. Ravidasvir C_{max} and AUC increased in a dose-proportional manner and 24-hour post dose plasma concentrations were above ravidasvir EC_{90} values for HCV genotypes 1 to 7. When dosed for 5 days, values of $AUC_{0-\infty}$ on day 1 and AUCτ on day 5 were similar.

Compared with healthy volunteers, subjects with chronic HCV infection seem to have higher inter-subject variability. Ravidasvir C_{max} and $AUC_{0-\infty}$ were generally lower in HCV subjects than in healthy volunteers, while T_{max} was similar in both groups.

Effects of food

When taken with food, the average C_{max} was reduced compared to the fasted condition, however the relative bioavailability was essentially unchanged (fed/fasted 103.4%). Median T_{max} was slightly delayed for the fed condition compared to the fasted condition (3.0 versus 2.0 hours). There is not expected to be a clinically relevant food effect with ravidasvir. Ravida can be taken with or without food.

Distribution

Ravidasvir is highly bound to plasma proteins, with an estimated fraction unbound of 1.9%, and is not widely distributed to and bound in extravascular tissues. The apparent volume of distribution (Vz/F) values were approximately 100 L at all dose levels and independent of the administered dose. *In vivo* studies in animals show that distribution to the liver, an important target organ for antiviral efficacy, is higher than to plasma.

Biotransformation

Metabolism in hepatic microsomes from all species was negligible, and oxidative metabolism of ravidasvir by human liver S9 fractions did not produce detectable glutathione conjugates. Mono-oxidation was the predominant metabolic pathway in all species studied, however, the pattern of metabolites varied between species. *In vivo* human studies indicated that ravidasvir was metabolically very stable, with only modest (~2%) metabolite formation detected. The most abundant metabolites in humans (M6 and M7, both of minor importance) resulted from mono-oxidation of ravidasvir.

Ravidasvir is not a substrate for the transport proteins OATP1B1, OATP1B3, or BCRP but is a substrate for P-gp. There is low likelihood that cytochrome P450 inhibitors or inducers would have a clinically meaningful effect on ravidasvir levels.

Elimination

Biliary excretion of ravidasvir appears to be the primary route of elimination of the absorbed dose, while renal excretion appears to be negligible. Following oral administration of ascending single doses in healthy volunteers, the relationship between $AUC_{0-\infty}$ and dose was linear indicating that the clearance for ravidasvir is independent of the administered dose.

In healthy volunteers, when 320 mg ravidasvir was administered once daily for 5 days, mean AUC0-∞ and AUC τ values were similar on Days 1 and 5, and resultant estimates for apparent clearance were nearly identical on Days 1 and 5, indicating that dosing for 5 days does not lead to either significant accumulation or induced elimination of ravidasvir. The pharmacokinetic profile supports dosing once a day.

Pharmacokinetic/pharmacodynamic relationship

HCV RNA levels were measured in chronic HCV-infected subjects dosed with 40 to 240 mg ravidasvir once daily for 3 days. HCV RNA levels rapidly declined from baseline in all cohorts with mean maximal reductions of 3.2, 3.5, 3.5 and 3.8 $\log_{10}$ IU/mL for the 40, 80, 160 mg and 240 mg doses, respectively. The HCV reductions for the 3 higher doses (3.5 to 3.8 $\log_{10}$ IU/mL) suggest that ravidasvir doses ≥80 mg once daily are near the top of the efficacy dose-response curve in HCV subjects. Plasma levels well above EC_{50} and EC_{90} were achieved with once daily dosing, leading to a reduction of viral load in all tested genotypes.

Linearity

The relationship of C_{max} and AUC to the dose appears linear in the dose range tested in healthy volunteers (80 mg to 320 mg) and HCV subjects (40 mg to 240 mg).

Pharmacokinetics in special populations

No specific pharmacokinetic studies have been conducted in special populations.

Race and gender

No clinically relevant pharmacokinetic differences due to race or gender have been identified for ravidasvir.

PRECLINICAL SAFETY DATA

In repeat toxicity studies, ravidasvir was well tolerated in the rat. Reversible effects of increased organ weight and ratios of adrenal glands, microvesicular vacuolation and diffuse hypertrophy of the zona fasciculata in the adrenal glands were observed. These were associated with increases in cholesterol, low-density lipoprotein and triglycerides. This was observed at the highest dose of 750 mg/kg/day. The no observed adverse effect level (NOAEL) of ravidasvir in rats was established at 300 mg/kg/day, at which AUC_{0-24} was approximately 7.6 times the clinical AUC_{0-24} at 200 mg. In the monkey, only mild fecal changes (discolored, loose, soft and mucoid stools) were observed at the dose of 500 mg/kg/day. The NOAEL of ravidasvir in monkeys was considered to be 500 mg/kg/day, at which AUC_{0-24} was 0.8 times and 1.4 times in males and females, respectively, the clinical AUC_{0-24} at 200 mg.

Ravidasvir was not genotoxic in a battery of *in vitro* or *in vivo* assays, including bacterial mutagenicity, chromosome aberration using human peripheral blood lymphocytes and *in vivo* rat micronucleus assays.

Carcinogenicity studies with ravidasvir have not been conducted. No evidence of mutagenic or clastogenic potential was observed in two *in vitro* tests (Ames bacterial mutagenicity, Mammalian Chromosomal Aberration Assay in human lymphocytes) or in an *in vivo* oral micronucleus study in the rat.

In animal reproduction studies, ravidasvir had no effects on fertility, embryofoetal development, lactation and postnatal development or sexual maturation at any dose (maximum 750 mg/kg/day) when administered orally to rats. However in rabbits, ravidasvir produced embryotoxicity and skeletal abnormalities at doses ≥100 mg/kg/day. Based on these teratogenic effects, the NOAEL for embryofoetal development in the rabbit was considered to be 25 mg/kg/day, at which the AUC_{last} was at least 1.7 times the clinical AUC_{0-24} at 200 mg. In view of these effects, Ravida is not recommended in pregnancy and, based on the possible excretion of ravidasvir in milk, breast-feeding is not recommended when taking Ravida.

Ravidasvir had no effects on the eyes or skin of pigmented rats in a definitive phototoxicity study at doses up to 750 mg/kg/day and is therefore considered not to be phototoxic.

STORAGE CONDITION

Store below 30°C.

Store the tablets in their original packaging.

SHELF LIFE

Product should not be used beyond the expiry date imprinted on the product packaging.

DOSAGE FORMS AND PACKAGING AVAILABLE

Ravida tablets are supplied in high density polyethylene (HDPE) bottle with a polypropylene child-resistant closure. Each bottle contains 28 tablets.

The following pack size is available: outer carton containing 1 bottle of 28 tablets.

PRODUCT REGISTRATION HOLDER

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